Tung-Cheng Wu

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Experience

- **DENSO Corporation [🌐]** *May 2026 - Present*
  Senior Research Engineer Tokyo, Japan
  - Led a PoCs team in designing action tokens for ADAS-based VLA model
- **DENSO International America, Inc. [🌐]** *Nov 2024 - May 2026*
  Senior Research Engineer Pittsburgh, PA, USA
  - Develop motion planning algorithm for autonomous driving system
- **Carnegie Mellon University [🌐]** *Nov 2024 - Apr 2026*
  Visiting Researcher Pittsburgh, PA, USA
  - **Research Topic**: Autonomous driving and uncertainty quantification
- **DENSO Corporation [🌐]** *Apr 2020 - Nov 2024*
  Research Engineer Tokyo, Japan
  - Develop a DNN-based algorithm for advanced driver assistance systems. This technology can forecast the lane-changing and cut-in intentions of surrounding vehicles early. It had conducted PoC and been transferred to the product team.
- **Industrial Technology Research Institute [🌐]** *Jul 2018 - Sep 2018*
  Intern Hsinchu, Taiwan
  - Develop a DQN-based navigation algorithm for quadcopter

Publications

< Submission >

1. **Tung-Cheng Wu**, Jeff Schneider (2027). **Surrogate Matters: Adaptive NPC Curriculum for Multi-Agent Autonomous Driving**. Manuscript submitted for publication in *The 2027 IEEE International Conference on Robotics and Automation (ICRA 2027)*.

< Conference >

1. **Tung-Cheng Wu**. (2027). **Is the Long Tail Really a Classifier Problem? A Domain-Conditional Re-examination of Decoupling under Foundation Models**. In *The IEEE/CVF Winter Conference on Applications of Computer Vision 2027 (WACV 2027)*, Florida, United States, Jan 2027.
2. **Tung-Cheng Wu**, Shau-Yin Tseng, Chin Feng Lai, Chia Yu Ho, Ying Hsun Lai. (2018). **Navigating Assistance System for Quadcopter with Deep Reinforcement Learning**. In *1st International Cognitive Cities Conference (IC3)*, Okinawa, Japan, July 2018, pp. 16-19. DOI: 10.1109/IC3.2018.00013
3. **Tung-Cheng Wu**, Jon-Fan Hu, Shu-Ling Peng. (2017). **Influences of the Matching Effects of Cognitive and Emotional Factors on Attitude Change**, In *Proceedings of the Annual Meeting of the Cognitive Science Society*, London, United Kingdom, July 2017, Vol.39, pp.3877.

< Journal >

1. Ying-Hsun Lai, **Tung-Cheng Wu**, Chin-Feng Lai, Laurence Tianruo Yang, Xiaokang Zhou. (2020). **Cognitive optimal-setting control of AIoT industrial applications with deep reinforcement learning**. In *IEEE Transactions on Industrial Informatics*, vol.17, no.3, pp.2116-2123, March 2021. DOI: 10.1109/TII.2020.2986501

Patent

< Provisional Patent > (undergoing Utility Patent Application)

1. **Tung-Cheng Wu**, Jeff Schneider (2026). **System and Method for Adaptive Surrogate Agent Curriculum for Multi-Agent Reinfordement Learning**.
2. **Tung-Cheng Wu**, Alexandre Capone, Jeff Schneider (2025). **Traffic Density and Episode Length-Based Curriculum Learning for Generative Driving Simulators**. Registration Date: Sep 2025.

Education

- **National Cheng Kung University** *Sep 2017 - Jun 2019*
  *Master of Science in Engineering Science, GPA: 4.1 / 4.3 (**best 0.7%**)* Tainan, Taiwan
  - **Thesis topic**: Design new updating strategy to improve deep reinforcement learning
  - **Major**: Path planning, Uncertainty quantification
- **National Cheng Kung University** *Sep 2013 - Jun 2017*
  Bachelor of Science in Psychology, GPA: 3.76 / 4.0 Tainan, Taiwan
  - **Major**: Computational cognitive neuroscience, Statistical learning theory

Skills

- **Programming Languages:** Python, Matlab, ROS, C++, R
- **Data Science & Machine Learning Toolkit:** PyTorch, TensorFlow, Keras, sk-learn
- **Other Tools & Technologies:** LaTeX, git
- **Research Skills:** Calibration, Data collection

Honors and Awards

- **Excellent Master Thesis** *Dec 2019*
  Institute of Information & Computing Machinery [🌐]
  - Excellent paper of CS theses in the same year from entire country
- **Excellent Paper Award** *Sep 2018*
  International Cognitive Cities Conference
  - One of four excellent papers
- **Scholar Award** *Jul 2018*
  Department of Engineering Science
  - Awarded to 5% best students across entire department

Additional Information

- **Languages:** Chinese (native), English (fluent), Japanese (JLPT N2)
- **Interests:** Swimming, Hiking, Real-Time Strategy (RTS) game

References

1. **Shinya Tanaka**
   Manager, Pittsburgh Innovation Lab
   DENSO International America, Inc.
   Email: shinya.tanaka@na.denso.com
   Relationship: Manager
2. **Jeff Schneider**
   Research Professor, School of Computer Science
   Carnegie Mellon University
   Email: jeff4@andrew.cmu.edu
   Relationship: Advisor during Visiting Researcher
3. **Ken Kinjo**
   Manager, AI R&I Div.
   DENSO Corporation
   Email: ken.kinjo.j3x@jp.denso.com
   Relationship: Manager
4. **Chin-Feng Lai**
   Professor, Engineering Science
   National Cheng Kung University
   Email: cinfon@ieee.org
   Relationship: Advisor during MS
5. **Jon-Fan Hu**
   Associate Professor, Psychology
   National Cheng Kung University
   Email: jfhu@mail.ncku.edu.tw
   Relationship: Advisor during BS